Antony Hubert

M.Sc. Mechanical Engineer — Metal Additive Manufacturing (PBF-LB/M)

+49 157 8344 2831 | antonyhubert03@gmail.com | linkedin.com/in/antony-hubert
Based in Aalen, Germany | Open to relocation worldwide
German Job Seeker Visa; eligible for EU Blue Card. Visa sponsorship welcome. Available immediately.

Publication DOI: 10.1016/j.procir.2024.08.060

PROFESSIONAL SUMMARY

Mechanical engineer specialising in Laser Powder Bed Fusion of Metals (PBF-LB/M, also L-PBF or SLM), with peer-reviewed results published in Procedia CIRP. I build process windows from first principles rather than by trial and error: Design of Experiments to map the parameter space, CAD and CFD to design the process hardware itself, and independent metallography, profilometry and SEM to verify what the process actually produced. Trained and published in Germany, with earlier industry experience in automotive CAD/CAE and CNC manufacturing in India. I am looking for the strongest technical environment available and am open to relocating anywhere that offers serious engineering work and the room to keep learning.

SELECTED ACHIEVEMENTS

Made a magnetically superior alloy manufacturable. FeSi6.5 offers up to 30% lower core losses than conventional electrical steel but is too brittle to roll or stamp, so industry cannot use it. I developed an in-process laser remelting method that cut areal surface roughness by approximately 60% (Sa 4.55 to 1.8 micrometres) at 0.152% porosity — smooth enough to carry the ultra-thin insulating layers that laminated motor and transformer cores require.

Turned grain structure into a process setting. Applying remelting cyclically rather than once converted columnar grains to a fully equiaxed structure through the full part height, with no cracking and no loss of dimensional accuracy. Remelting frequency became the control variable — a dial an engineer can set, not an outcome to be accepted.

Improved the machine, not just the parameters. Designed, simulated, built and commissioned an inlet gas circulation system with build platform heating for a PBF-LB/M research machine — the full component lifecycle on operating research hardware.

CORE COMPETENCIES

Additive Manufacturing: PBF-LB/M (L-PBF, SLM), DMLS, in-process laser remelting, laser parameter and scan-path optimisation, volumetric energy density modelling, melt-regime control (conduction and keyhole), porosity and defect analysis, support structure design, Design for Additive Manufacturing (DfAM), metal powder handling, inert atmosphere control

Simulation and Analysis: ANSYS Fluent (CFD), ANSYS Mechanical and APDL (FEA), Design of Experiments (DoE), Taguchi method (L9 orthogonal array), ANOVA, regression analysis, Minitab

CAD and Design: CATIA V5, Siemens NX, PTC Creo, AutoCAD, reverse engineering, GD&T (ISO 1101), Body-in-White structural design, EV component layout

Materials Characterisation: Metallography (sectioning, mounting, grinding, polishing, etching), optical microscopy, Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), areal profilometry (KEYENCE VR-3100), tactile roughness metrology (MarSurf M400), porosity quantification, micro-CT, grain morphology analysis

Software and Systems: Autodesk Netfabb, Autodesk Machine Control Framework (AMCF), Materialise Magics, ZEISS ZEN Core, Minitab, Python (foundation), machine learning fundamentals

Manufacturing and Delivery: 4-axis CNC machining and CAM, toolpath optimisation, process qualification, technical documentation, R&D project coordination, training and mentoring

PROFESSIONAL EXPERIENCE

Academic Research Assistant — PBF-LB/M Process Development — 09/2023 – 03/2024

Hochschule Aalen, LaserApplication Center (LAZ), Aalen, Germany — SmartPro / Smart-ADD research programme

- Owned the FeSi6.5 process development end to end, from powder to a documented, reproducible parameter window; results published in Procedia CIRP (2024).
- Ran parameter campaigns across three experimental phases and more than 30 combinations, spanning 78–208 J/mm³ volumetric energy density.
- Separated two effects the field routinely conflates: molten-metal edge deposition scales with power intensity, while remelting depth scales with areal energy intensity. Controlling them independently is what made a stable window achievable.
- Performed all characterisation personally — metallography, optical microscopy to 500×, SEM, areal profilometry and tactile roughness metrology — with no external lab dependency.
- Completed the inlet gas circulation system begun as a working student — added build platform heating for grain-structure control, manufactured, integrated and commissioned it on the ALC2. It then carried the entire thesis campaign and the resulting publication.

- Coordinated planning and data exchange with external research partners; delivered every project milestone on schedule.

Working Student — Metal Additive Manufacturing **05/2023 – 08/2023**
Hochschule Aalen, LaserApplication Center (LAZ), Aalen, Germany

- Began the inlet gas circulation system for the ALC2 research machine: requirements, CATIA V5 nozzle and diffuser geometry, and ANSYS Fluent flow analysis to eliminate the recirculation zones where spatter settles back onto the powder bed. Carried through to manufacture and commissioning in the research assistant role that followed.
- Operated PBF-LB/M systems across the full workflow: CAD, CFD, build preparation, machine operation, post-processing and material evaluation.
- Reduced failed builds through improved support strategy and build-parameter selection on complex metallic geometries.

Application Engineer — CAD / CAE **03/2022 – 09/2022**
CADD Centre Training Services Pvt. Ltd., Chennai, India

- Specialised in automotive Body-in-White structural design, EV component layout, GD&T and reverse engineering.
- Trained and mentored over 100 engineers and students across CATIA V5, Siemens NX, PTC Creo, ANSYS and AutoCAD.
- Provided design consultation and manufacturing-feasibility review on client projects.

CAD / CAM Engineer — CNC Manufacturing **06/2021 – 02/2022**
Axis Automation Pvt. Ltd., Chennai, India

- Programmed and operated 4-axis CNC machining centres for precision automotive components.
- Reproduced complex legacy parts from physical samples by reverse engineering.
- Optimised toolpath strategies (feed rate, depth of cut, tool selection), raising throughput and reducing scrap.

EDUCATION

M.Sc. Advanced Materials and Manufacturing **10/2022 – 09/2025**
Hochschule Aalen (Aalen University), Faculty of Mechanical Engineering and Materials Technology, Germany

- Final grade 2.7 (German scale: 1.0 excellent, 4.0 pass). Focus: metal additive manufacturing, powder metallurgy, materials characterisation, soft magnetic alloys.
- Thesis: Optimization of Layer Surface Topography in FeSi6.5 Soft Magnetic Components Through In-Process Laser Remelting in PBF-LB/M. Supervisors: Prof. Dr. Harald Riegel, Prof. Dr. Dagmar Goll.

B.E. Mechanical Engineering **07/2017 – 08/2021**
Jeppiaar Institute of Technology, Chennai, India — CGPA 8.5 / 10

- Thesis (lead author): porosity minimisation in DMLS cobalt-chromium for medical implants. A Taguchi L9 array over laser power, scan speed and hatch spacing identified scan speed as the dominant factor and reached 2.8% porosity against a 27.06% worst case, in 9 runs instead of 27.

PUBLICATION

Hofele M., Antony H., Kolb D., Riegel H. (2024). Production of soft magnetic FeSi6.5 parts with precise layer topography using integrated laser remelting at PBF-LB/M to induce multi-material lamination. Procedia CIRP, Volume 124, pages 6–11. DOI: 10.1016/j.procir.2024.08.060

CONTINUING PROFESSIONAL DEVELOPMENT

Independent Study — AI and Machine Learning for Manufacturing (2025 – present). Self-directed work in machine learning, data analysis and Python, aimed at in-process monitoring, defect detection and quality prediction — the direct continuation of the real-time melt-pool monitoring identified as future work in my thesis.

CERTIFICATIONS

- Applied Ergonomics — NPTEL / IIT Madras, 2020
- CAE and CFD using ANSYS — CIPET, Chennai, 2019
- CAD/CAM using PTC Creo — CIPET, Chennai, 2019
- CAD/CAM using CATIA V5 — CIPET, Chennai, 2019

LANGUAGES

- English — C1 (Advanced). Language of my Master's study, thesis and publication.
- German — B1 (Intermediate); B2 examination scheduled before end of December 2026.
- Tamil — C2 (Native / Bilingual).